

Epigenetic ageing clocks: statistical methods and emerging computational challenges

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Abstract

Over the past decade, epigenetic clocks have emerged as powerful machine learning tools, not only to estimate chronological and biological age but also to assess the efficacy of anti-ageing, cellular rejuvenation and disease-preventive interventions. However, many computational and statistical challenges remain that limit our understanding, interpretation and application of epigenetic clocks. Here, we review these computational challenges, focusing on interpretation, cell-type heterogeneity and emerging single-cell methods, aiming to provide guidelines for the rigorous construction of interpretable epigenetic clocks at cell-type and single-cell resolution.

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Introduction

Biological age aims to quantify the physiological condition of an organism and is distinct from chronological age in being more informative of health or lifespan¹. Biological age can thus vary considerably between individuals of the same chronological age. In its most generic form, biological age is hard to measure, as it represents an umbrella term that comprises a myriad of different manifestations of ‘unhealthy ageing’, which may include, for example, high blood pressure and cognitive decline. These manifestations are often organ-specific and quantifiable. Together, they determine the risk of mortality of an individual, which could be viewed as the ultimate quantifiable end point of biological age. Importantly, biological age is also malleable, that is, it is modifiable by specific interventions such as caloric restriction, diet or exercise, which have been shown to extend health and lifespan in animal models and humans^{2,3}.

One of the most profound biological insights of the past few decades has been the demonstration that various forms of biological age are quantifiable with molecular ageing clocks^{4,5} and that these clocks can also be used to evaluate the efficacy of anti-ageing interventions^{4–10}. Molecular ageing clocks are machine learning algorithms trained from specific molecular omics data, including genomic, epigenomic, transcriptomic, proteomic, glycomic and metabolomic biomarkers⁶. The most established ageing clocks are epigenetic clocks^{4,11}, which use DNA methylation (DNAm) features at CpG dinucleotide pairs¹² to approximate chronological age^{4,13–15} or various forms of biological age^{5,11,16–19} (Fig. 1a).

DNAm clocks stand out for multiple reasons. First, DNAm can be measured with very high accuracy in a genome-wide manner in many samples using Illumina bead array technology^{20,21}, with more cost-effective targeted approaches, for example, pyrosequencing²², digital droplet PCR²² and TIME-seq²³, offering increased scalability without compromising robustness or accuracy. Second, DNAm clocks are the most accurate for chronological age prediction; for example, the Zhang clock can predict the chronological age of a blood sample with a median absolute error of only ± 1 –3 years¹⁴. This accuracy is important for specific applications such as forensics²⁴. The accuracy is remarkable given that most age-associated DNAm changes are small, typically only involving a 2–5% DNAm change over many decades²⁵, even when assessed in sorted cell populations²⁶. Third, DNAm clocks are universally applicable across the mammalian kingdom²⁷, which offers the potential to better understand the molecular determinants of the maximum lifespan of a species^{27–29} and widens their application to fields such as wildlife conservation³⁰. Fourth, DNAm clocks can quantify biological age at different hierarchies and scales, from predicting morbidity and all-cause mortality in humans^{17–19,31}, to measuring different forms of cellular age such as telomere length^{32,33} or stem cell divisions^{34–37}. They have also been associated with a wide range of diseases and disorders, including obesity³⁸, HIV infection³⁹, Down syndrome⁴⁰, neurodegeneration⁴¹ and type 2 diabetes mellitus⁴², as well as with early menopause⁴³ and pregnancy⁴⁴. Finally, DNAm clocks have undergone extensive independent validations⁶.

Mathematically, DNAm clocks yield estimates of biological age acceleration, defined as the relative increase or decrease of the age estimate of a clock, called DNAm age, relative to chronological age (Fig. 1b). Intuitively, the more accurate a clock is for chronological age (for example, the Zhang clock¹⁴), the less informative it can be with respect to quantifiable biological age measures such as mortality risk (Fig. 1a). Conversely, a clock more informative of mortality risk (for example, GrimAge¹⁹) cannot be very predictive of chronological age, although

some correlation would be expected since age itself is a major risk factor (Fig. 1a). This duality between chronological and biological age has spurred the development of next-generation DNAm clocks trained on non-epigenetic ageing biomarkers (for example, inflammatory and metabolic markers in blood plasma, such as levels of C-reactive protein (CRP) or telomere length), or on disease-promoting exposures (for example, smoking pack-years), either at a single time point (PhenoAge and GrimAge clocks)^{17,19,32} or, more recently, from successive longitudinal measurements of ageing biomarkers aimed at capturing the pace of ageing (DunedinPoAm and DunedinPACE clocks)^{45,46} (Table 1). The ability to quantify biological age molecularly in terms of the accumulation of DNAm changes relative to a suitable ground state is of particular interest to the anti-ageing field because epigenetic reprogramming has already been shown to reverse cellular ageing, rejuvenating cells and organs^{7–9,47,48}. Thus, DNAm clocks have also emerged as useful tools to quantify the effect of health-promoting and cellular rejuvenating interventions^{47,49,50}, and are especially powerful in well-controlled animal models such as the mouse^{9,11,51,52}.

Despite the epigenetic clock field blossoming into an active and promising area of ageing research, major experimental, technological and computational challenges remain that limit our understanding and practical application of these clocks. Among these challenges is the elucidation of the intricate network of cellular and molecular mechanisms that underpin ageing clocks. The current view is that most DNAm clocks reflect composites of many underlying processes, including DNA repair^{9,53}, DNA damage^{53,54}, cell division^{34,55}, cell-type composition⁵⁶, somatic mutations⁵⁷, circadian rhythm^{58–62} and metabolic stress^{63,64} (Fig. 1c). An emerging computational challenge is therefore how to dissect the clocks in terms of these underlying processes. In addition, the question arises of how to build epigenetic clocks in a manner that directly aids interpretation. A closely related challenge is posed by cell-type heterogeneity^{65,66}, as most epigenetic clocks have, by and large, been trained from bulk tissue samples, which are composed of many different cell types. Hence, current DNAm age estimates constitute averages that are not necessarily informative of which cell types in a tissue are ageing faster or slower. Thus, there is an urgent need to dissect such DNAm age estimates into the various cell-type-specific components. Another challenge is causality: although experimental work in mouse models has shown that the ‘ticking’ of the clock at the DNAm level could reflect a causal process that contributes to ageing and biological ageing⁹, whether computational approaches can help establish causality in human cohorts remains uncertain.

This Review discusses these challenges from a computational and statistical perspective. The aim is to raise awareness of the key limitations of current clocks and to describe computational methods to help resolve outstanding challenges. These guidelines may serve as pointers to computational and experimental work needed for the epigenetic clock field to move forward.

Defining biological age acceleration

In the context of epigenetic clocks, two main definitions of biological age acceleration have been used⁵: one is defined by the difference between DNAm age and chronological age, known as absolute age acceleration (AAA), whilst the other is defined by the residuals of a linear regression of DNAm age against chronological age, known as relative age acceleration (RAA) (Fig. 2a). In this section, we discuss the reasoning behind each measure as well as cases in which each measure might be preferable.

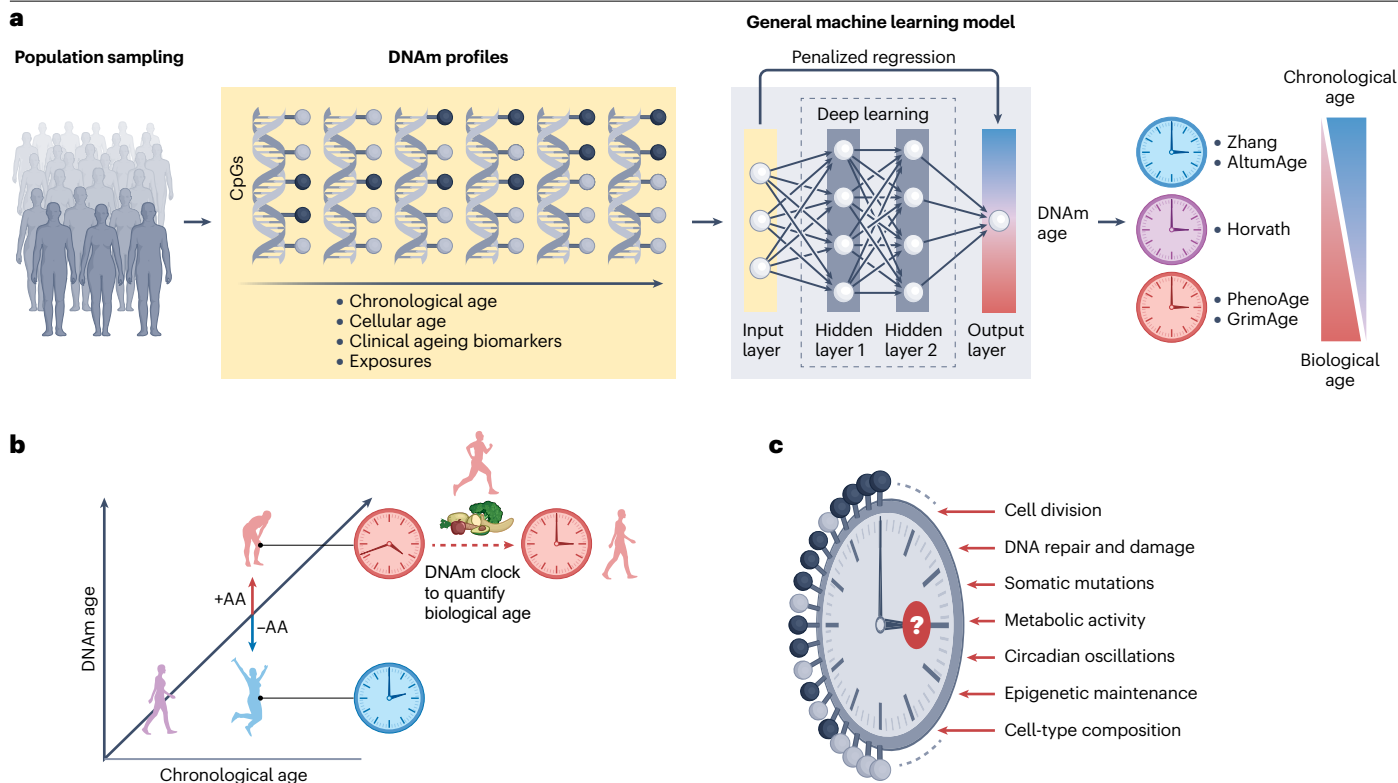


Fig. 1 | Biological age and epigenetic clocks. a, Construction of an epigenetic clock. The input consists of genome-wide DNA methylation (DNAm) data and a variable to be estimated, which could be chronological age, various quantifiable forms of biological age (for example, mortality risk), cellular age (for example, telomere length or mitotic age), clinical ageing biomarkers (for example, levels of C-reactive protein) or exposures (for example, smoking pack-years). Machine learning methods can include penalized linear regression or deep learning. The resulting epigenetic clocks yield an estimate of DNAm age. Of note, clocks more predictable of chronological age are less predictive of the inter-individual variation in biological age, with the converse also true, although a clock highly informative of biological age should still correlate with chronological age. Examples of current epigenetic clocks and how they fare in predicting chronological versus biological age are shown. **b**, The meaning and relativity of

biological age, its quantification via DNAm age and age acceleration (AA) with epigenetic clocks, and how it is malleable by (for example, health-promoting) interventions. The regression line represents the expected relationship between DNAm age and chronological age. Points above the regression line indicate positive age acceleration (+AA), whereas points below the line indicate negative age deceleration (–AA). Individuals who exhibit +AA are older than their peers at the same chronological age, whereas individuals with –AA are younger than their peers at the same chronological age. DNAm clocks provide a readout of innate processes that accompany ageing. +AA is associated with age-related diseases and functional decline but interventions such as dietary changes and exercise have shown potential in ‘turning back the clock’. **c**, Proposed mechanisms and processes that can drive age-associated DNAm changes at clock CpGs.

Ideally, datasets used for training, validation and subsequent application should be as similar as possible, in the sense that they should all have been generated with the same technology, under the same laboratory conditions and on independent subsamples of the same underlying population. In such an ideal setting, the resulting value of an epigenetic clock should be directly interpretable as chronological or biological age. Thus, if the predicted value differs markedly from the true value, the clock is not useful. However, in practice, there could be a plethora of technical reasons why predictions are not aligned (in an absolute sense) with expected values (Fig. 2b). For instance, several clock CpGs might be missing in an external dataset because they did not pass quality control or because it was generated with a different technology. Hence, some individuals who are ageing at a normal rate could be predicted to be ageing faster when using the absolute definition of age acceleration, AAA (Fig. 2b). Incidentally, this erroneous prediction would also result if defining AAA by comparison to the fitted regression line from the training set. This simple example illustrates

why one may consider the alternative residual-based definition of age acceleration, RAA, since taking residuals of a regression fit can adjust for these and other biases (Fig. 2b). A notable statistical reason why RAA is preferable is that, by mathematical construction, RAA always displays zero correlation with chronological age, whereas AAA is likely to be negatively correlated (Fig. 2c): it follows from ordinary or penalized least squares regression that the variance of the predictions is always less than that of the explained variable (age), which leads to a flattening of the regression line, the implication being that older and younger samples are more likely to display negative and positive AAA, respectively (Fig. 2c). This phenomenon is similar albeit not entirely equivalent to the ‘regression to the mean effect’ coined by Galton^{67,68}.

Should we always use RAA? Unfortunately, the answer is not straightforward and depends on both the goals of the analysis and the biological or clinical characteristics of the external dataset. For instance, AAA can be evaluated in a single sample irrespective of other measurements, whereas RAA depends on the measurements of

other samples. As a concrete example, consider the Horvath pan-tissue clock, which has been shown to display age acceleration in liver samples from individuals with obesity compared to control individuals with a normal body mass index (BMI)³⁸. If we apply the Horvath clock to a cohort that only includes samples from people with clinical obesity, the RAA would not necessarily reflect a correlation with BMI because all samples have a very high BMI (Fig. 2d). By contrast, and assuming that technical differences are minor, AAA would correctly predict age-acceleration values above zero for most, if not all, patients (Fig. 2d). However, if the external dataset contains both individuals with normal weight as well as individuals with clinical obesity, RAA is more

appropriate because the residuals reflect the considerable variation in BMI (Fig. 2d). Conversely, if one were to fit the regression line through the control samples only, AAA would correctly predict acceleration in individuals with obesity. Yet, another twist occurs if the individuals with clinical obesity happen to be significantly younger than those with normal BMI. In this third scenario, fitting a regression line through this population would fit the average of younger individuals with obesity to the average of older healthy individuals, with the ensuing regression line appearing flatter, and with the residuals failing to capture the age-acceleration effect of obesity, in effect because BMI is confounded with age itself (Fig. 2d). Using AAA instead would correctly predict age

Table 1 | DNA methylation clocks for humans

Name	Tissue	Programme	Method	Web links	Refs.
Chronological age clocks					
Horvath	Multi-tissue	R	Elastic net	https://DNAMage.clockfoundation.org/	4
Hannum	Whole blood	R	Elastic net	https://www.bioconductor.org/packages/release/bioc/html/methylclock	15
PCHorvath1/2 PCHannum	Multi-tissue, whole blood	R	PC-based regression	https://github.com/MorganLevineLab/PC-Clocks/	71
Zhang	Whole blood	R	Elastic net	https://www.bioconductor.org/packages/release/bioc/html/methylclock	14
SkinBlood	Skin, whole blood	R	Elastic net	https://DNAMage.clockfoundation.org/	75
AltumAge	Multi-tissue	Python	Deep neural net	https://github.com/rsinghlab/AltumAge	88
IntrinClock	Multi-tissue	R	Elastic net	https://doi.org/10.5281/zenodo.10426597	198
MEAT	Skeletal muscle	R	Elastic net	https://bioconductor.org/packages/release/bioc/html/MEAT.html	141
Cortical clock	Brain cortex	R	Elastic net	https://github.com/gemmashireby/CorticalClock	140
PedBE	Buccal swabs	R	Elastic net	https://github.com/kobor-lab/Public-Scripts/	199
Biological age clocks					
PhenoAge	Whole blood	R	Elastic net	https://www.bioconductor.org/packages/release/bioc/html/methylclock	17
PCPhenoAge	Whole blood	R	PC-based regression	https://github.com/MorganLevineLab/PC-Clocks/	71
GrimAge & GrimAge2	Whole blood	R	Elastic net	https://DNAMage.clockfoundation.org/	18,19
PCGrimAge	Whole blood	R	PC-based regression	https://github.com/MorganLevineLab/PC-Clocks/	71
DunedinPoAm	Whole blood	R	Elastic net	https://github.com/dan-belsky/DunedinPoAm38	46
DunedinPACE	Whole blood	R	Elastic net	https://github.com/danbelsky/DunedinPACE	45
PRC2-AgeIndex	Multi-tissue	Python	Deviation counter	https://github.com/moqri/PRC2-AgeIndex	155
Mitotic clocks					
epiTOC1/pcgtAge	All tissues	R	Deviation counter	https://github.com/aet21/EpiMitClocks	34
HypoClock	All tissues	R	Deviation counter	https://github.com/aet21/EpiMitClocks	35
epiTOC2	All tissues	R	Model-based counter	https://github.com/aet21/EpiMitClocks	37
epiCMT	Blood	R	Deviation counter	https://github.com/Duran-FerrerM/Pan-B-cell-methylome	200
RepliTali	All tissues	R	Elastic net	https://github.com/aet21/EpiMitClocks	55
stemTOC	All tissues	R	Deviation counter	https://github.com/aet21/EpiMitClocks	160
Telomere length clock					
DNAmTL	Whole blood	R	Elastic net	https://www.bioconductor.org/packages/release/bioc/html/methylclock	32
Causality clocks					
DamAge & AdaptAge	Whole blood	R	Elastic net	https://github.com/albert-ying/ClockBasis	167

For additional epigenetic clocks as well as software packages that implement a large number of these we refer the reader to the methylclock R package²⁰¹ and the pyaging Python package²⁰². PC, principal component.

acceleration in individuals with obesity. Thus, AAA can be preferable over RAA even though the latter measure will be advantageous in most applications involving a population of individuals such as a clinical trial or an epidemiological cohort study.

Statistical versus biological variation

Another challenge relates to the statistical significance of biological age-acceleration estimates (Fig. 2e). Even one of the most accurate epigenetic clocks for chronological age, the Zhang clock, which was trained on more than 13,000 whole-blood samples, is imperfect¹⁴. Thus, an unresolved question is whether an observed deviation is a sampling error of the algorithm or a biological effect that reflects an individual that is ageing genuinely faster or slower than the average value expected for a peer of the same age in a normal ageing population (Fig. 2e). Currently, no published epigenetic clock offers a statistical guideline as to which deviation threshold signals the onset of significant age acceleration or deceleration. However, in the context of clinical or preclinical decision-making, this threshold is important to be able

to inform individuals on whether their biological age estimates are within the range of healthy ageing or not⁶⁹. Estimating the sampling error of a clock, which will depend on the sample size of the training cohort¹⁴, is thus critical. In the case of chronological ageing clocks, this statistical challenge is compounded by the fact that these clocks have been trained from very large cohorts that will inevitably also include technical and biological outliers (for example, blood samples may include leukaemia cases). Consequently, training clocks in such cohorts could lead to inflated measures of sampling error. To mitigate the impact of severe outliers, an iterative approach could be considered, whereby outliers are progressively removed to refine a subpopulation suitable for model training, and from which a 95% confidence interval for the error distribution can be derived. For any individual from an independent dataset, a personalized probability that its deviation is reflective of age acceleration or deceleration could then be computed (Fig. 2e). Although not fully realizing this goal, a recent study developed a Bayesian model framework⁷⁰, training it on blood samples from a large cohort to obtain individual-specific posterior estimates of age

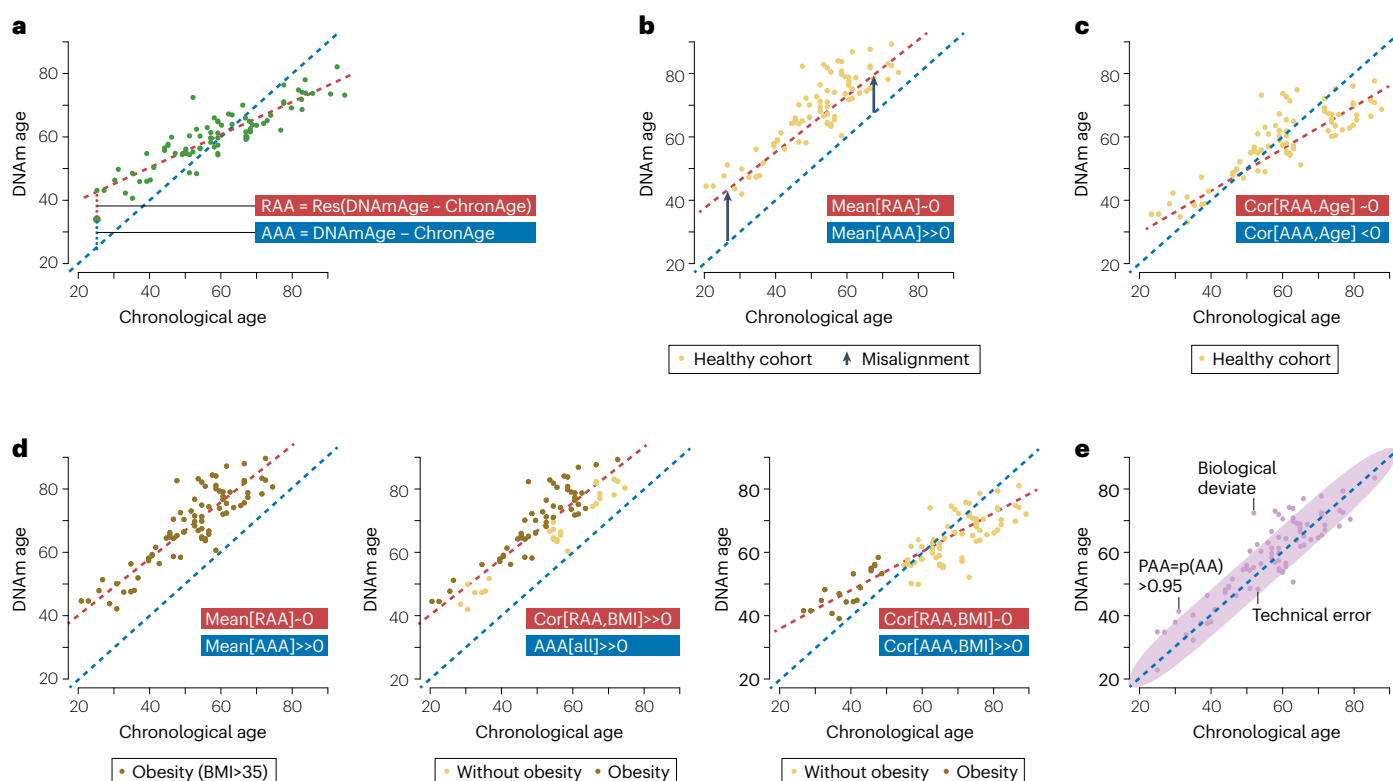


Fig. 2 | Mathematical definitions of biological age acceleration and their properties. **a**, Biological age acceleration can be defined in two ways. One is via the difference between DNAm age (the estimate from the epigenetic clock) and chronological age, called absolute age acceleration (AAA). Another way is via the residuals of a linear regression fit of DNAm age versus chronological age, called relative age acceleration (RAA). **b**, Owing to uncontrollable factors (for example, missing clock CpGs, different DNAm technology, normalization or different tissue types), DNAm age estimates obtained in an average healthy cohort could be misaligned. According to the AAA, samples from healthy individuals would be incorrectly predicted to be ageing faster, whilst the RAA would correctly predict a mean age acceleration of approximately zero. **c**, Illustration of the regression to mean effect, whereby the variance of the predictions (DNAm age) is always less than that of the explained variable (age),

leading to a flattening of the regression fit and a negative correlation (Cor) of AAA with chronological age. By construction, RAA displays a zero correlation with chronological age. **d**, Scatter plots of DNAm age versus chronological age for three different hypothetical scenarios: samples from a cohort of individuals with clinical obesity (body mass index (BMI) > 35) (left); a cohort of individuals with obesity and age-matched individuals with a normal BMI (middle); and a cohort of young individuals with obesity and older individuals with a normal BMI (right). In the first and third scenarios, AAA works better than RAA, whilst RAA is more appropriate in the second scenario. **e**, If epigenetic clocks are to be informative for each individual, it would be important to quantify a 95% confidence interval, so that deviations within the interval are indicative of average ageing, whilst deviations outside indicate a high probability that the corresponding individuals are ageing faster or slower than the average. PAA, probability of age acceleration.

acceleration, a framework that could be extended to yield personalized probabilities. At the population level, age-acceleration measures are often correlated to epidemiological factors or health outcomes. When comparing epigenetic clocks to each other in their ability to predict such health outcomes, it is important to ensure that age-acceleration estimates are standardized by their variance to ensure that hazard ratios are indeed comparable, as previously described^{45,71}.

Statistical paradigms for epigenetic clock construction

Epigenetic clocks have been built using different statistical paradigms. We provide a brief overview of the most important ones here and refer readers to Box 1 for additional technical details.

Penalized linear regression and principal component analysis-based models

Most epigenetic clocks, including Horvath, Hannum, PhenoAge, GrimAge and DunedinPACE, have been built with a penalized multivariate linear regression framework called elastic net^{5,72} (Table 1 and Box 1). In this model, DNAm values of specific CpGs in a sample are linearly combined with CpG-specific weights to yield the age estimate of a clock. The specific CpGs and their weights are inferred from a training set using a penalized optimization function and cross-validation that favours less complex models containing fewer CpGs, which is critical to avoid overfitting⁷³. Of note, because of the multivariate nature of these models, the final optimized model can include CpGs that are not strongly correlated with age across the training set samples⁷⁴. Whilst this reflects the nature of multivariate optimization to seek complex combinations of CpGs that model residual unexplained variation in specific sample subsets, it can be easily avoided by preselecting CpGs that correlate with age in univariate analysis^{15,75}.

Despite their popularity, penalized linear regression models display several limitations. First, naive application of these models assumes a global linear relation between DNAm and age, whilst individual CpGs often display only an approximately linear relationship, with deviations to linearity occurring primarily during childhood, adolescence (<20 years)⁴ and at older age (>70 years)⁷⁶. Although non-linearities can be captured by using a non-linear transformed version of age⁴, which is then regressed linearly to the DNAm features (Box 1), this procedure still requires specification of the boundaries where non-linear behaviour arises. Of note, most elastic net models are trained on cohorts where younger and older samples are under-represented, which explains why the resulting accuracy of a clock is much lower in these younger and older age groups. Thus, penalized linear regression models are limited if the aim is to capture more complex and unknown non-linear patterns, as modelling these would necessitate explicit non-linear methods such as Gaussian processes⁷⁷, probabilistic kernel density-based methods⁷⁸ or deep learning (discussed below).

Another limitation of the penalized linear regression framework was exposed by a study comparing PhenoAge clock DNAm age estimates of technical replicates, which reported significant deviations on the order of 9–10 years⁷¹, thus raising concerns about the robustness of these estimates. The same study also proposed a specific solution, demonstrating how using inferred principal components, in place of individual CpGs, can reduce CpG-level noise⁷¹ (Box 1). Thus, principal component versions of Horvath, Hannum, PhenoAge and GrimAge clocks were built (Table 1) and shown to improve reproducibility of technical replicates⁷¹. Whilst the principal component versions of most clocks were found to be more accurate in predicting mortality risk than

their respective original versions, the principal component version of GrimAge was not, probably because GrimAge already includes principal components associated with plasma proteins and smoking pack-years, or simply because of sampling variability, indicating that more comprehensive analyses are needed to assess the advantages of principal component-based clocks. Another shortcoming of penalized regression has been the limited biological insight gained from them, which is somewhat surprising given that linear models are generally easier to interpret than more complex non-linear models such as random forests⁷⁹. For instance, whilst functional enrichment analysis has shown that Horvath clock CpGs with positive and negative weights are often located near the binding sites of the Polycomb-repressive complex 2 (PRC2) and enhancer regions, respectively⁴, the specific role of the clock CpGs remains unclear. One notable exception is a CpG mapping to the *ELOVL2* gene, which encodes an enzyme whose activity regulates ageing in the mouse retina^{80–82}. It is therefore plausible that the rigid linear framework of penalized regression may obscure biological interpretation if the most informative patterns involve complex non-linear interactions.

Deep learning models

Deep learning⁸³ has emerged as a powerful machine learning paradigm to capture complex, non-linear, predictive data patterns, which has revolutionized much of science, including biology^{84–86} (Box 1). Deep learning has been most successful when applied to complex correlative data patterns in a spatial (for example, amino acid sequence and image recognition) or temporal (for example, speech recognition and text prediction) domain, where it can automatically detect predictive complex feature representations, often at multiple hierarchical levels of abstraction. However, in the context of epigenetic clocks, deep learning has so far only led to rather modest and debatable improvements in predictive accuracy^{87–90}. One likely reason is that current deep learning implementations treat CpGs as structureless features without consideration of their relative spatial positions in the genome. Taking into account the genomic context of CpGs is therefore critical. Doing so would also reduce the effective dimension of feature space, as current DNAm assays typically measure up to one or several million CpGs, which is two to three orders of magnitude bigger than the largest available sample sizes (10,000 to 100,000 samples). However, as far as the prediction of chronological age is concerned, deep learning is unlikely to result in substantially improved predictions because current penalized linear regression-based clocks are already close to the limit of theoretical accuracy¹⁴. Nevertheless, adopting a deep learning framework could lead to an improved prediction of biological age or to an improved understanding of epigenetic ageing provided it is used in conjunction with explainable artificial intelligence (XAI) tools^{91,92}. When applied to epigenetic clocks, XAI tools yield importance scores for the underlying CpGs and can identify potential CpG pairs or complex CpG combinations whose non-linear interactions contribute significantly to the predictive accuracy. These importance scores can also be sample specific, enabling identification of CpGs that are predictive for one sample but not another, which can thus help address the challenge posed by inter-individual heterogeneity. As a concrete example, AltumAge is an epigenetic clock based on a deep neural network (Table 1), which, in conjunction with a popular XAI tool called SHAP⁹³, allows ranking of CpGs in order of importance, revealing a strong enrichment of CTCF-binding sites among top-ranked CpGs⁸⁸. Graph convolutional neural networks^{94–97} represent another promising deep learning paradigm for epigenetic clock construction,

Box 1 | Statistical frameworks for epigenetic clock construction

Penalized linear regression

By far the most popular statistical framework for epigenetic clock construction has been penalized multivariate regression⁷², where a given function F of age (A), is expressed as a linear weighted sum of the CpG DNA methylation (DNAm) values. The aim is to identify the subset of CpGs and their regression weights that best predict $F(A)$ in a least-squares sense. Regularization, the introduction of a penalty term to penalize models that include more CpGs, is conducted to avoid overfitting. The penalized log-likelihood that one aims to maximize is typically of the form:

$$\log L(w, \lambda) = - \sum_s (F(A_s) - w_0 - \sum_c w_c \beta_{cs})^2 - \lambda \left(\alpha \sum_c |w_c| + (1 - \alpha) \sum_c w_c^2 \right)$$

where c labels the CpGs, s the samples, β_{cs} are the corresponding DNAm beta-values, and $0 \leq \alpha \leq 1$ specifies the form of the penalty function ($\alpha = 1$ is called lasso penalty, $\alpha = 0$ defines ridge-regression and intermediate values, usually $\alpha = 0.5$, is referred to as ‘elastic net’)⁷². The maximization of $\log L(w, \lambda)$ involves inference of the optimal weights w and the penalty parameter λ , which is accomplished using a cross-validation strategy⁷³. The clock is then specified by the optimal λ value (λ^*) and the estimated weights $\hat{w}_c$ at that optimal λ^* value. Typically, only a few hundred CpGs will have non-zero weights and contribute to the predictor. For a new sample s with a DNAm profile defined by β_{cs} , the clock estimate is then $F(A)_s = \hat{w}_0 + \sum_c \hat{w}_c \beta_{cs}$. If there are good reasons to assume that the relation between DNAm and age is linear, one can choose $F(A) = A$. However, F could be a non-linear function of age, in which case we refer to it as transformed age. For instance, F could be piecewise linear, with linear functions of different slopes defined for different age ranges. In that case, the estimated $F(A)_s$ value needs to be transformed back to an age estimate $\hat{A}_s$. Most epigenetic clocks are elastic net predictors^{4,14,17}. For instance, the Horvath clock⁴ is an elastic net predictor with $\alpha = 0.5$, and with $F(A)$ a non-linear function of age composed of two linear functions, one for ages less than 20 years and another for all ages greater than or equal to 20 years. This was done to accommodate the empirical observation that DNAm changes are more rapid during the first 20 years of life.

PCA-based clocks

The rationale behind these clocks is to reduce technical noise by first applying a principal components analysis (PCA) to dimensionally reduce a DNAm data matrix and then to train an Elastic Net regression model in the latent embedded space⁷¹. Thus, in this framework, the DNAm beta-values in the log-likelihood function get replaced by the inferred loads of the principal components, and the summation is no longer over CpGs c but over the principal components p . Mathematically, if the training set X is decomposed in a PCA as $X = UV$, then one aims to maximize

$$\log L(w, \lambda) = - \sum_s (F(A_s) - w_0 - \sum_p w_p V_{ps})^2 - \lambda \left(\alpha \sum_p |w_p| + (1 - \alpha) \sum_p w_p^2 \right)$$

where V_{ps} is the load of component p in sample s . Because not all principal components may carry significant variance, random matrix theory²⁰³ or permutation²⁰⁴, could be used to estimate the number of significant components and only sum over those. However, biologically relevant components could carry ultra-low variance^{71,205}; therefore, it is recommended to apply the elastic net on all available principal components since noisy components would be eliminated anyway. Thus, the number of principal components would typically be the number of samples, as the number of CpGs is much greater. PCA-based clocks have been shown to reduce inter-replicate variation⁷¹ and have also been applied to other types of data, for example, clinical data²⁰⁶. Interestingly, supervised versions of PCA, where PCA is applied to features that correlate with the phenotype of interest, have shown improvements over unsupervised PCA in other contexts²⁰⁷ but remain underexplored in the epigenetic clock field.

Deep learning-based clocks

Deep learning is an advanced machine learning paradigm that draws on the power of ‘big data’ to learn complex and hierarchical non-linear feature representations of data, typically encoded in the form of multiple hidden layers, and which can be implemented in either an unsupervised or supervised context^{83–85}. The traditional deep learning algorithm is based on the multi-layer neural network, where layers are organized hierarchically, with each layer composed of a fairly large number of units (‘neurons’), each unit taking as input a weighted linear combination of outputs from the previous layer and transforming it according to some non-linear activation function. Mathematically, the output of unit i in layer l , x_{il} , can be expressed as $x_{il} = f(\sum_h w_{hi} x_{hl})$, where w_{hi} label the weights between units h and i in layers $l-1$ and l , and where f is the non-linear activation function, for example, $f(x) = \text{ReLU}(x) = \max(x, 0)$ or $f(x) = 1/(1 + e^{-x})$. The weights and other parameters that may enter the non-linear activation functions can be learned using the backward propagation algorithm²⁰⁸, which recursively applies the chain rule of derivatives to identify the parameter changes that optimize an objective function, usually the least square error between prediction and true value. Two of the most popular and successful deep learning algorithms are convolutional and recurrent neural networks⁸³. A convolutional neural network is a deep, multi-layer neural network specifically designed for processing array or grid-based data such as images, which can detect and process predictive spatial patterns at multiple length scales. Recurrent neural networks are another form of deep neural network, specifically designed for sequential (temporal) data, that has found applications in speech recognition and language processing. So far, epigenetic clocks have been built with the traditional deep multi-layer neural network^{87,88,193}, where the use of multiple hidden layers may identify complex non-linear interactions between CpGs that are highly informative of age or biological age. If subsequently interpreted with an explainable artificial intelligence tool^{91,92,209}, a deep learning algorithm could lead to novel biological insights. For instance, in the context of a multi-layer neural network, explainable artificial intelligence tools, such as SHAP⁹³ and DeepPINK²¹⁰, allow features, such as CpGs, to be ranked according to their importance in the predictive model.

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Noise barometer clocks

A fourth important paradigm for building epigenetic clocks, albeit only for quantifying biological age, relies on the notion that biological ageing can be measured from DNAm outliers^{74,100,101}. These ‘outlier CpGs’ would display ultra-stable DNAm values in a healthy population but outlier DNAm values in individuals who age faster than normal, for instance, because of exposure to disease risk factors. Observations of such outlier CpGs in normal tissue at risk of cancer indicate that (1) they are relatively infrequent, and hence ‘stochastic’, events, and (2) that the number of such outliers in a given sample is a measure of cancer risk^{100,101}. In the context of large blood-based epigenome-wide association studies, the number of such DNAm outliers has been termed the ‘noise barometer’⁷⁴ or ‘stochastic epimutation load’^{105,108}. These stochastic outlier CpGs do display age-associated DNAm changes but are not selected or favoured

by standard machine learning methods (for example, penalized regression) because the latter favour CpGs that display more linear and homogenous age-associated DNAm changes⁷⁴. By this, we mean that the DNAm variance across samples of the same age group remains small and does not change with age. In contrast, outlier CpGs would be characterized by heterogeneous age-related DNAm changes, typically displaying an increased DNAm variance with age. Thus, these outlier CpGs can be identified with statistical tests for differential variability⁶⁵, for instance, the Bartlett^{100,211} or Levene tests^{211,212} for binary phenotypes (exposure versus non-exposure) or the Breusch–Pagan test²¹³ for continuous phenotypes such as age¹⁰⁹. In the context of epigenetic ageing, the Breusch–Pagan test would assess if the squared residuals from a linear regression of DNAm against age are themselves correlated with age.

where prior biological information, such as biological pathways, regulatory or protein interactions, can be incorporated in the form of graphs and integrated with DNAm data to help biological interpretability. For instance, an XAI tool, such as PGExplainer⁹⁸, could be applied on top of the graph convolutional neural network to rank edges in the graph according to their importance in predicting the outcome (for example, chronological age or biological age). Of note, a graph convolutional neural network has already been used for cancer classification⁹⁹.

Noise barometer clocks

Another important paradigm for building epigenetic clocks aimed at biological age prediction is based on the premise that CpGs most informative of biological age display ultra-stable DNAm levels in healthy individuals, possibly reflecting homeostatic vital loci, but outlier DNAm levels in individuals experiencing biological age acceleration, for instance, because of exposure to disease risk factors^{74,100,101} (Box 1). Thus, according to this paradigm, the most informative CpGs for biological age are not those that correlate most linearly with chronological age.

This paradigm has been particularly successful in the context of cancer-risk prediction where the DNAm outliers in normal preneoplastic tissues likely reflect subclonal expansions at higher risk of cancer development as demonstrated in the context of cervical cancer¹⁰⁰ and breast cancer¹⁰¹. These DNAm outliers constitute infrequent events and thus appear largely stochastic^{100,101}. In the context of blood tissue, such DNAm outliers, also called stochastic epimutations, increase with age¹⁰² and likely reflect genetic clonal mosaicism, which has independently been shown to increase with age and to be predictive of cancer risk^{103–105}. Clonal haematopoiesis itself has also been shown to be associated with epigenetic age acceleration¹⁰⁶, although the correlation is weak, consistent with CpGs undergoing stochastic epimutations generally not being part of epigenetic clocks¹⁰⁷. In general, accumulation of stochastic DNAm outliers defines a ‘tochastic epimutation load’ that could be viewed as a general ‘noise barometer’ that is predictive of biological age⁷⁴; however, at present, it remains unclear which diseases they could be informative of, beyond cancer^{103–105} and Parkinson disease¹⁰⁸. Nevertheless, these studies highlight the fact that CpGs can display very different forms of age-related DNAm change and that those displaying increased variance and stochasticity with age are not those best suited for predicting chronological age^{109,110}. Thus, if the aim is

to build clocks for biological age, focusing on differentially variable CpGs might be more fruitful (Box 1). This line of investigation should be pursued more vigorously in the future.

Epigenetic clocks and cell-type heterogeneity

Challenges posed by cell-type heterogeneity

To date, epigenetic clocks have been trained from bulk tissue samples, which are composed of many different cell types. This cell-type heterogeneity poses several different challenges. First, cell-type proportions within a tissue can change with age (Fig. 3). For instance, age-related lymphoid-to-myeloid skewing is a well-known phenomenon in blood¹¹¹, as is a shift from naive to mature T cell fractions^{112–115}. In brain, inhibitory and excitatory neurons decrease and increase with age, respectively^{116,117}. Thus, training a clock in such tissues would select for CpGs that are discriminatory of these cell subtypes, with their DNAm levels in the individual cell types of the tissue not necessarily changing with age. Even for a pan-tissue clock, such as the Horvath clock, which was trained on multiple tissue types, cell-type heterogeneity could lead to distorted DNAm age estimates⁵⁶ because the training set underlying the Horvath clock was predominantly whole blood, and many of the other, solid tissues used in training, such as lung or liver, are infiltrated by a considerable amount of immune cells¹¹⁸ that display similar age-related shifts to those observed in whole blood¹¹⁹. Confirming this finding, different immune cell subsets (for example, naive and effector CD8⁺ T cells) from the same donor were shown to display widely different DNAm age estimates⁵⁶. Nevertheless, a substantial component of epigenetic clocks is not due to changes in cell-type composition, as these clocks can predict chronological age in sorted immune cell datasets (for example, naive CD4⁺ T cells from the BLUE-PRINT consortium¹²⁰) with many of the underlying clock CpGs also changing with age simultaneously across different immune cell subsets⁸². Thus, current epigenetic clocks should be viewed as composites, with an ‘extrinsic’ component that reflects age-related shifts in cell-type composition and an ‘intrinsic’ component that reflects the ageing of individual cell types (Fig. 3).

The composite nature of epigenetic clocks thus raises several outstanding questions that could be addressed by advanced cell-type deconvolution methods^{121–123}. For instance, how much of the predictive accuracy of current epigenetic clocks is driven by cell-type

heterogeneity and how much of it is cell-intrinsic? Do cell types within a tissue display the same biological age or do they age at different rates, and are these differential rates of ageing related to specific exposures? To what extent is the physiological age of a tissue determined by the biological ages of its constituent cell types? Dissecting the composite nature of DNAm clocks is also critical for longitudinal studies in human populations^{45,46,124}, as repeated measurements in blood tissue from the same individual are subject to substantial variations in immune-cell-type composition that could be influenced by a plethora of environmental confounders, including time of sampling, infections, sleep, exercise, jet lag, stress or diet. Hence, future work should address at what time scales we can dissect longitudinal intrinsic changes from extrinsic ones, and which ones are more informative of future health outcomes.

A posteriori adjustment for cell-type heterogeneity

The traditional way of addressing the cell-type heterogeneity challenge has been to conduct an a posteriori correction of DNAm age estimates using the estimated cell-type fractions obtained from

the joint application of a cell-type deconvolution algorithm and a tissue-specific DNAm reference matrix^{65,121,123,125}. For instance, if using a residual-based definition for age acceleration, one can adjust for the heterogeneity of the tissue by including the estimated cell-type fractions as additional covariates in the linear regression model, leading to the notion of intrinsic age acceleration⁵. This contrasts with the residuals obtained by simply regressing DNAm age against chronological age, referred to previously as RAA but also widely known as extrinsic age acceleration¹²⁶. However, a posteriori correction via intrinsic age acceleration is not an ideal method to adjust for cell-type heterogeneity because cell-type-specific CpGs still contribute to the DNAm age value of a clock⁵⁶, and because the obtained age acceleration only represents an average over all underlying cell types, thus being uninformative of cell-type-specific biological ageing.

Explicit adjustment for cell-type heterogeneity

The cell-type heterogeneity challenge would be better addressed by adjusting for it during the training of the epigenetic clock, which could,

Epigenetic clocks built from bulk tissue are composites

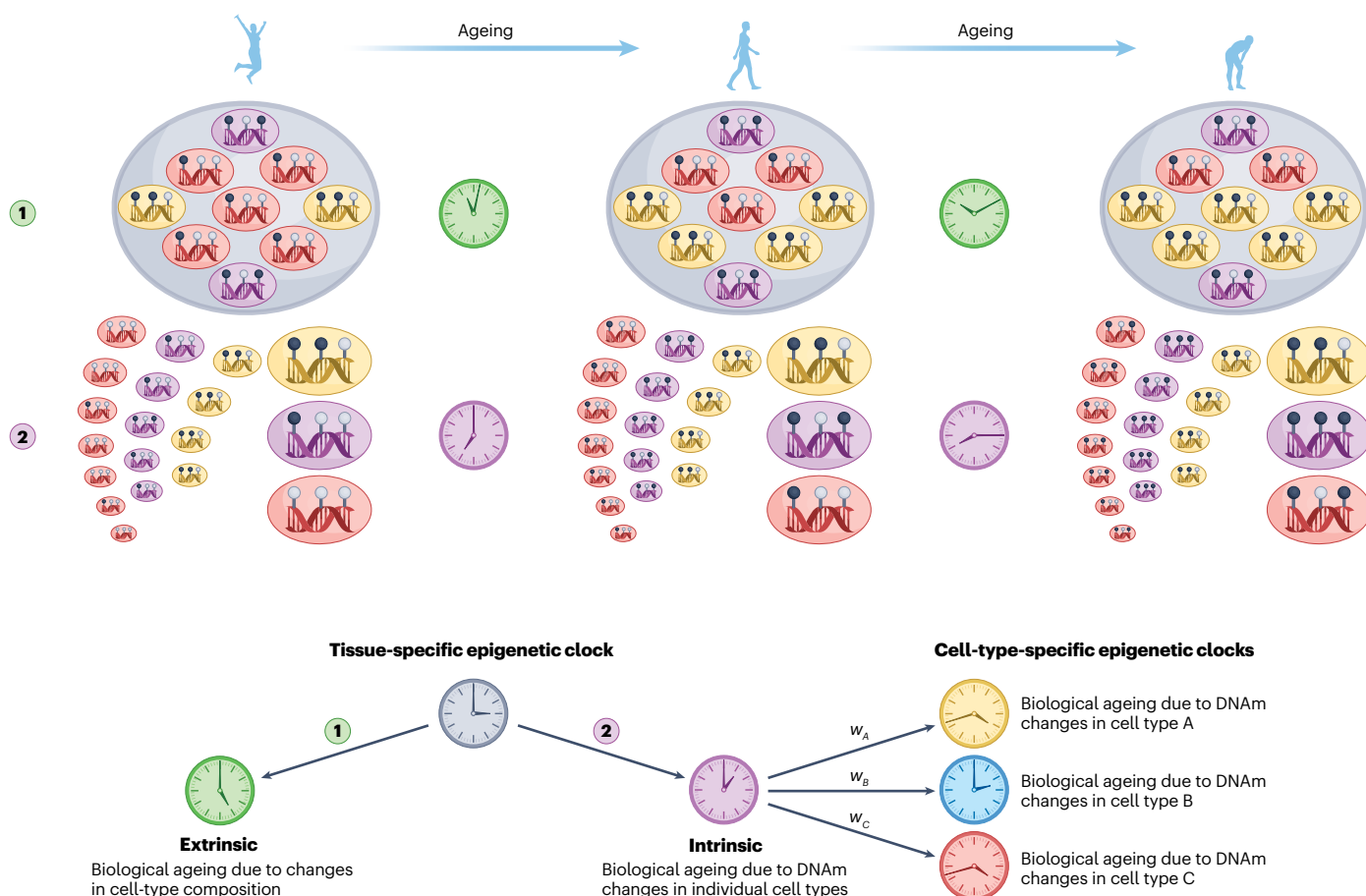


Fig. 3 | Cell-type heterogeneity and the composite nature of epigenetic clocks. Biological ageing of a tissue has two components, one reflecting the age-related change in cell-type composition of the tissue, and another reflecting the ageing (here depicted by age-related DNA methylation (DNAm) changes) of each cell type in the tissue. Epigenetic clocks trained from bulk tissue samples are thus composites of two underlying clocks, with one ‘extrinsic’ clock measuring the age-related change in cell-type composition (clock 1) and another ‘intrinsic’

one that measures the biological ageing of individual cell types in the tissue (clock 2). The latter would be a weighted average of the DNAm ages of individual cell types (for example, cell types A, B and C), with weights determined by the underlying cell-type proportions so that $w_A + w_B + w_C = 1$. To measure biological age of individual cell types requires the construction of cell-type-specific epigenetic clocks (Box 2).

in principle, be addressed experimentally. For instance, one could generate DNAm profiles of sorted cell populations, and whilst such data are available for specific immune^{120,127} and epithelial cell subtypes (for example, bronchial epithelial cells)¹²⁸, it is clearly cumbersome and cost-prohibitive to generate for all cell types within a tissue and for the many samples needed to comprehensively capture age-associated DNAm changes. Generating DNAm data at single-cell resolution is equally cost-prohibitive, limited by technologies that only generate very sparse data^{129–131}, and has so far been restricted to profiling cells from only a few donors, not allowing for rigorous epigenetic clock construction¹¹⁶. Alternative cost-effective solutions are computational and could include the pre-selection of CpGs that are not cell-type-specific when analysing bulk tissue data²². However, this approach assumes that DNAm levels of cell-type-specific CpGs do not change with age, which is unrealistic. Another solution would involve cell-type deconvolution of the bulk tissue DNAm dataset used in training, so that the clock explicitly adjusts for the underlying variations in cell-type fractions. We note that the estimation of such cell-type fractions, traditionally restricted to easily accessible tissues such as blood, is now possible for a wide range of different tissue types, thanks to the construction of corresponding DNAm reference atlases^{123,132,133}. Although the estimation of such cell-type fractions is still limited to a resolution of around 5–20 cell types, with an estimation error that can vary from approximately 5–30% depending on tissue and cell type, such atlases nevertheless enable the construction of tissue-specific epigenetic clocks that are adjusted for cell-type heterogeneity.

Tissue-specific epigenetic clocks

Most current epigenetic clocks are tissue specific but have not been adjusted for cell-type heterogeneity. For instance, Hannum¹⁵, Zhang¹⁴, PhenoAge¹⁷ or GrimAge¹⁹ are blood specific because they have been built from blood tissue datasets (Table 1). Constructing tissue-specific clocks for other tissue types is an important endeavour for the more accurate quantification of biological age as it is not reasonable to assume that all tissues and organs age at the same rate^{134–137}. Although pan-tissue clocks, such as the Horvath clock, have been used to demonstrate that specific organs, including breast and cerebellum, age faster^{136,138} and slower¹³⁷ compared with other tissue types, it is still plausible that these differential rates can be attributed to variations in cell-type composition. Moreover, whilst initial studies indicated that epigenetic clocks trained exclusively or predominantly in blood (which includes the Horvath pan-tissue clock) could be informative of biological ageing in less accessible tissues such as brain¹³⁹ or liver³⁸, more recent studies have highlighted the inadequacy of these clocks and advocated for the need to develop tissue-specific clocks, as conducted for brain¹⁴⁰ and skeletal muscle¹⁴¹. To build tissue-specific clocks adjusted for cell-type heterogeneity, one could first adjust the DNAm data for variations in cell-type fractions, subsequently training a penalized regression model⁷² on the residuals. The ensuing clock would then reflect a prediction of chronological age that is entirely based on age-associated DNAm changes that occur in individual cell types and is much less likely to track age-related changes in cell-type composition, at least not at the resolution of the defined fractions. Whilst this approach can help dissect and quantify the relative contributions of extrinsic and intrinsic ageing underlying an epigenetic clock (Fig. 3), it still only yields an average DNAm age and thus remains uninformative of the biological ageing rates of individual cell types.

Cell-type-specific epigenetic clocks

The procedure described above would result in a DNAm age estimate that is a weighted average over the DNAm ages of the individual cell types making up a tissue, with the weights proportional to the corresponding cell-type fractions in the sample. Thus, this clock does not necessarily lead to biological age estimates that are free from confounding by cell-type heterogeneity: for instance, hepatocytes in the liver of individuals with obesity might age faster than the surrounding liver stromal cells but a clock built as described above would only yield a biological age averaged over hepatocytes and stromal cells and would therefore be confounded by the relative fractions of these two cell types. This limitation applies to the construction of the clock itself, since using residuals only ensures that the DNAm age is not driven by changes in cell-type composition. Thus, an emerging computational challenge is how to build cell type-specific epigenetic clocks, and subsequently how to infer cell-type-specific DNAm ages when applied to a bulk tissue sample (Box 2).

One possible approach, assuming a sufficiently large bulk tissue training dataset is available, would be to apply an algorithm, such as CellDMC¹²², designed to identify cell-type-specific differential DNAm, in this instance to detect cell-type-specific age-related differentially methylated cytosines (age-DMCs). Once identified, one could then apply a machine learning model (for example, penalized regression) restricting to the cell-type-specific age-DMCs (Box 2). Since the training of the clock is on CpGs that change with age in the given cell type, this approach should yield cell-type-specific DNAm age estimates.

An alternative way would be to infer the underlying data tensor, a three-dimensional array with dimensions defined by CpGs, cell types and samples (Box 2). Projecting the estimated tensor into the dimension specified by each cell type results in a cell-type-specific data matrix defined over CpGs and samples, which is already naturally adjusted for cell-type heterogeneity. This would then allow direct construction of cell-type-specific clocks using the same machine learning models as before (Box 1). This tensor approach has been successful in the context of bulk tissue gene expression data¹⁴² and similar attempts on DNAm data have indicated it might work here too¹⁴³, although not without controversy¹⁴⁴. Indeed, the inference of the data tensor is technically challenging as it requires large datasets, and existing methods have not been extensively tested and validated¹⁴⁴.

At present it is also unclear how many age-DMCs are truly cell-type-specific, which is an important prerequisite to justify building cell-type-specific clocks. Because most of the DNA methylome changes with age, with the direction of change largely dictated by local CpG density and the level of DNAm in a suitably defined ground state (for example, fetal or newborn stage)¹⁴⁵, most age-DMCs seem to be shared between cell types⁸². Age-DMCs are more likely to differ between cell types whenever their underlying ground-state DNAm values in these cell types are distinct. It follows that selection of CpGs based on cell-type-specific ground-state DNAm values prior to training might substantially aid the construction of cell-type-specific clocks.

Single-cell clocks

Although the most direct solution to building cell-type-specific clocks would be to start from single-cell data, the high cost, sparseness and unscalability of single-cell DNAm assays^{129,130} has hindered such work. Nevertheless, attempts have been made to build cell-type-specific clocks from bulk tissue samples that can yield DNAm age estimates in single cells representing the dominant cell type in the tissue, and which could be used to test for age acceleration or deceleration

Box 2 | Computational methods for building cell-type-specific epigenetic clocks

Intrinsic and semi-intrinsic cell-type-specific clocks

Given an appropriately powered bulk tissue DNA methylation (DNAm) data set X defined over CpGs and samples, the underlying cell-type fractions $\hat{f}_{ts}$ for each sample s and cell type t are first estimated using a tissue-specific DNAm reference matrix. Such a reference matrix can be derived from resources such as EpiDISH¹²⁵ or the EpiSCORE DNAm atlas¹²³. Alternatively, it can be built de novo if appropriate DNAm data of sorted cells¹³² or single cells¹¹⁶ are available. Once the proportions of all main cell types in a tissue have been estimated, one can then apply an algorithm, such as CellDMC¹²², to infer age-associated DNAm changes in each cell type. This must be done in a training set X_{train} , so it is important that there are independent model selection X_{modsel} and validation sets X_{val} available to avoid overfitting. For cell types with sufficient age-related differentially methylated cytosines (age-DMCs), we then proceed by training machine learning predictors M (for example, penalized elastic net regression) parameterized by a complexity or penalty parameter λ , restricting the training to the age-DMCs found in the given cell type of interest. Each of these trained models is then evaluated in the model selection set using a metric such as root-mean-square-error (RMSE) or median absolute error. The optimal model minimizing the RMSE (or median absolute error) and defined by parameter λ^* is declared the true predictor. A separate blinded validation set is then used to validate the selection of the optimal model, that is, one would like to see a similar RMSE in the model selection and validation sets. In training the clock, there is a choice of whether to perform this on the original beta-valued DNAm data matrix X or on the residuals R obtained by regressing out the effect of cell-type fractions. The former might be preferable if cell-type fractions change with age because, in such a scenario, adjusting for cell-type heterogeneity could inadvertently remove a biological signal. However, in both cases, the training is performed on the same set of age-DMCs, that is, on CpGs that change with age in the given cell type, and so both procedures lead to a cell-type-specific clock that quantifies intrinsic ageing. In the former case, we refer to the cell-type-specific clock as ‘intrinsic’; in the latter case, where residuals are not used, as ‘semi-intrinsic’. Pseudocode for the construction of the intrinsic clock is given below:

Intrinsic cell-type-specific clock:

$$X = (X_{\text{train}}, X_{\text{modsel}}, X_{\text{val}}) \Rightarrow \hat{f}_{ts} \forall t, s$$

$$(X_{\text{train}}, \hat{f}_{ts}) \Rightarrow \text{ageDMCs} \forall t$$

in single cells following an intervention¹⁴⁶. As a concrete example, one study used reduced-representation bisulfite-sequencing data of 30 mouse liver tissue samples encompassing a wide age range to build an approximate hepatocyte clock, which was subsequently validated in single hepatocyte cells from young and old mice¹⁴⁶. Although this study is encouraging, it is worth noting that the non-hepatocyte component in liver is as high as 40–50%^{123,147}, the implication being that any clock built from liver tissue without adjustment for the underlying cell-type heterogeneity is likely to still be confounded. Hence, any subsequent validation in single-cell data should be interpreted with great caution, especially if these only involved a fairly small

$$\text{Im}(X_{\text{train}} \sim \hat{f}_{ts}) \Rightarrow \text{residuals}(R_{\text{train}})$$

$$(R_{\text{train}}, \text{ageDMCs}(t), \lambda) \Rightarrow M(\lambda, \text{ageDMCs}(t)) \text{ for range of } \lambda \text{ values}$$

$$M(\lambda, \text{ageDMCs}(t)) \Rightarrow R_{\text{modsel}} \Rightarrow \text{RMSE}(\lambda) \Rightarrow \text{optimal } \lambda^* \Rightarrow M^*(\text{ageDMCs}(t))$$

$$M^*(\text{ageDMCs}(t)) \Rightarrow R_{\text{val}} \Rightarrow \text{validation}$$

Tensor-based cell-type-specific clocks

An alternative strategy to the construction of cell-type-specific epigenetic clocks would be to first infer from X the underlying data tensor defined over CpGs c , cell types t and samples s . In other words, we decompose the data matrix X_{cs} as:

$$X_{cs} = \sum_{t=1}^K \hat{f}_{ts} T_{cts}$$

to infer the tensor T_{cts} . Of note, this tensor inference procedure only requires the estimated cell-type fractions $\hat{f}_{ts}$, which are estimated as before. As such, it is blinded to the phenotype of interest (in our case, age). The inference of the tensor could be accomplished using adaptations of tools such as CIBERSORTx, which have been successfully applied in the context of bulk RNA-sequencing data¹⁴². Once the tensor has been inferred, it is clear that T_{cts} defines, for each cell-type t , an ordinary DNAm data matrix $Z_{cs}^{(t)} = T_{cts}$ defined over CpGs c and samples s . These data matrices are naturally adjusted for cell-type heterogeneity since they are defined at the resolution of each cell type. Subsequently, the machine learning model M is directly applied to $Z_{cs}^{(t)}$ with $s \in \text{TrainingSet}$ to learn a range of models parameterized by λ . The model selection set, that is, $Z_{cs}^{(t)}$ with $s \in \text{ModSelSet}$, would then be used to select the optimal model λ^* and the validation set $Z_{cs}^{(t)}$ with $s \in \text{ValidationSet}$ to verify that the chosen model is not overfitted. In pseudocode:

Tensor-based cell-type-specific clock:

1. $X = (X_{\text{train}}, X_{\text{modsel}}, X_{\text{val}}) \Rightarrow \hat{f}_{ts} \forall t, s$
2. $(X, \hat{f}_{ts}) \Rightarrow T_{cts} \Rightarrow Z_{cs}^{(t)} = T_{cts}$
3. $(Z_{cs}^{(t)} \text{ s.e. train}, \lambda) \Rightarrow M(\lambda)$ for range of λ values
4. $M(\lambda) \Rightarrow Z_{cs}^{(t)} \text{ s.e. modsel} \Rightarrow \text{RMSE}(\lambda) \Rightarrow \text{optimal } \lambda^* \Rightarrow M^*$
5. $M^* \Rightarrow Z_{cs}^{(t)} \text{ s.e. val} \Rightarrow \text{validation}$

number of cells or samples. A specific computational challenge when applying a clock to single-cell DNAm data is the ultra-high sparsity of such data, which prevents application of the clock due to a potentially large fraction of missing CpGs. However, a solution is to adopt a probabilistic framework that allows for different sets of CpGs to be used for different cells¹⁴⁶. Alternatively, one could use an imputation approach where clock CpGs with missing values in a given single cell are replaced by highly correlated ‘surrogate’ CpGs that have values in the given cell¹⁴⁸. Care should be taken when inferring such surrogate CpGs: for instance, if derived by computing correlations across bulk tissue samples, adjustment for cell-type heterogeneity is necessary because,

otherwise, the correlations might only reflect covariation in cell-type composition.

As the number of single-cell DNAm studies grows, it is helpful to observe the lessons gained from a single-cell RNA sequencing (scRNA-seq) clock study¹⁴⁹. Given the inherent noise of single-cell data, how best to summarize clock age estimates over single cells remains unclear. As shown by the study, the development of a cell-type-specific clock from scRNA-seq data benefits from considering ‘bags’ of single cells called ‘meta-cells’ to achieve an optimal trade-off between reducing noise (achieved by using fewer meta-cells, each composed of many single cells) and increasing sample size (achieved by using more meta-cells, each composed of fewer single cells)¹⁴⁹; there seems to be an optimal bag size of meta-cells, which may be influenced by study-specific factors, but which, for the scRNA-seq data considered in ref. 149, was typically around 15–20 cells per meta-cell. Another related statistical challenge is how best to derive the cell-type-specific biological age of a donor from the different meta-cells or cells in that donor. For instance, is the estimated biological age variation over meta-cells of importance, or only the average, or would taking an upper 95% quantile be more biologically relevant? Taking such an upper quantile over single cells or meta-cells might help identify a rare subpopulation of cells that are ‘disease initiators’. Alternatively, would we fare better by simply turning the single-cell data into cell-type-specific pseudo-bulk profiles, in which case the single-cell data are only useful as a means of generating purified cell population-level data? These are very pertinent questions that will need to be addressed as the epigenetic field moves forward, embracing the latest advances in single-cell technology.

Epigenetic clock mechanisms and causality

The construction of most epigenetic clocks is blind to the actual molecular mechanisms that give rise to the age-related DNAm changes. For instance, in the case of the telomere DNAm clock³², although trained to predict measured telomere length, the actual mechanisms that relate specific DNAm changes to telomere length are unclear. In the case of other clocks, such as Horvath or PhenoAge, some of the underlying CpGs are markers of naive and mature T cell subsets but, otherwise, the underlying mechanisms underpinning these clocks remain unknown. Computational¹⁵⁰ and experimental¹⁵¹ approaches have helped to determine potential molecular mechanisms.

Computational elucidation of epigenetic clock mechanisms

Among computational attempts to help elucidate the mechanistic underpinnings of epigenetic clocks is a spectral clustering method that was applied to seven large DNAm datasets encompassing multiple tissue types¹⁵⁰. Clustering was performed on around 5,000 CpGs derived from various clocks, revealing 12 distinct correlative modules that seem to track different biological ageing processes. For instance, some of these modules displayed age reversal upon cellular reprogramming, a strong correlation with future all-cause mortality but weak acceleration in tumours, whilst other modules, strongly enriched among mitotic clocks^{34,35,37,55,160}, were found to display the exact opposite pattern, being strongly accelerated in tumours, consistent with them tracking cell division. Interestingly, and except for the Zhang clock, most non-mitotic clocks, including Horvath, PhenoAge, GrimAge and DunedinPoAm, were composed of four to six modules, suggesting that these clocks are composites of many underlying mechanisms. By contrast, the module composition of mitotic clocks and the Zhang clock was relatively homogeneous with, for example, epiTOC1 (ref. 34) and the Zhang clock each being composed of only two modules, which were also entirely

distinct from each other. The study also highlighted similar module compositions for GrimAge and DunedinPoAm¹⁵⁰, consistent with these two clocks being those most correlated with all-cause mortality.

Mitotic clocks as mechanism-specific clocks

The observation that mitotic clocks were composed of only two correlative modules¹⁵⁰ is consistent with these clocks having been trained by restricting to CpGs thought to track one particular process: cell proliferation. For instance, epiTOC1/2 and stemTOC (Table 1) were built by restricting to specific CpGs universally unmethylated in fetal tissue and marked by the PRC2, for which there is substantial biological evidence that DNAm gains at these sites track the number of stem cell divisions^{152–155} (Fig. 4a). EpiTOC2, in particular, was built from an explicit mathematical recurrence equation that models DNAm transmission between successive cell generations¹⁵⁶, allowing direct estimation of stem cell division rates and numbers that display good correlations with published estimates³⁷. Two other mitotic clocks, HypoClock and RepliTali (Table 1), were built from isolated CpGs that occur in a WCGW (W = A/T) context (hence, called solo-CpGs) that are found enriched in partially methylated domains^{35,55} (Fig. 4a). These solo-CpGs lose DNAm upon cell division^{35,55}, probably because of incomplete DNAm maintenance by DNA methyltransferase 1 (DNMT1) in late-replicating, partially methylated domain regions³⁵. Thus, mitotic clocks constitute one of the few examples of epigenetic clocks that are mechanism-specific, which aids downstream interpretation. Indeed, the mitotic age of a tissue, representing the cumulative number of stem cell divisions, has been proposed to be a major cancer risk factor^{157–159}. In line with this hypothesis, epigenetic mitotic clocks aimed at predicting mitotic age^{34,35,37,55,160} display universal age acceleration in cancer and precancerous lesions, holding promise as cancer-risk prediction tools^{37,157,159,160}. Of note, age acceleration in cancer and precancerous lesions is not universally observed for non-mitotic clocks³⁴. However, a study performed in mice showed that pan-tissue mouse^{161,162} and pan-mammalian DNAm clocks²⁷ do display an increased acceleration rate in the stem cell pools of more proliferative tissues, such as colon or blood, compared with quiescent stem cells such as satellite cells in skeletal muscle¹⁶³. Hence, it is plausible that a relatively minor component of pan-tissue human clocks is also registering changes in cell proliferation. Future experimental and computational work will be necessary to further resolve and quantify this.

Stochasticity underpinning epigenetic clocks

Several computational studies have shed light on whether the mechanisms underlying epigenetic clocks could be stochastic or not^{164–166}. Age-associated DNAm changes accrue at genomic loci within single cells in a quasi-stochastic fashion, with some loci displaying a preferential propensity for changes that otherwise seem to be largely stochastic, with a directionality of change that is largely dictated by local CpG density and DNAm level at birth²⁶. For example, PRC2 sites that are constitutively unmethylated in fetal tissue display preferential hypermethylation with age, with a probability or propensity for methylation gain that is variable between such PRC2 sites¹⁵³. Although it should be noted that any stochastic temporal counting process in single cells would naturally lead to a linear predictor of cumulative counts at the cell population level¹⁶⁴, quantifying the degree of stochasticity of epigenetic clocks could provide important clues into the nature of the underlying epigenetic changes¹⁶⁵. Markov chain simulation models showed that the more predictive a clock is for chronological age (for example, the Zhang clock), the more likely it is to be driven by a purely

stochastic process, whereas deviations from stochasticity are more likely to reflect biological ageing¹⁶⁵. For instance, by replacing the Phe-noAge clock, which displays age acceleration in the blood of smokers¹⁷,

with a stochastic version of it, age acceleration in smokers was no longer observed, suggesting that biological age acceleration associated with smoking is driven by non-stochastic processes¹⁶⁵. By contrast, stochastic

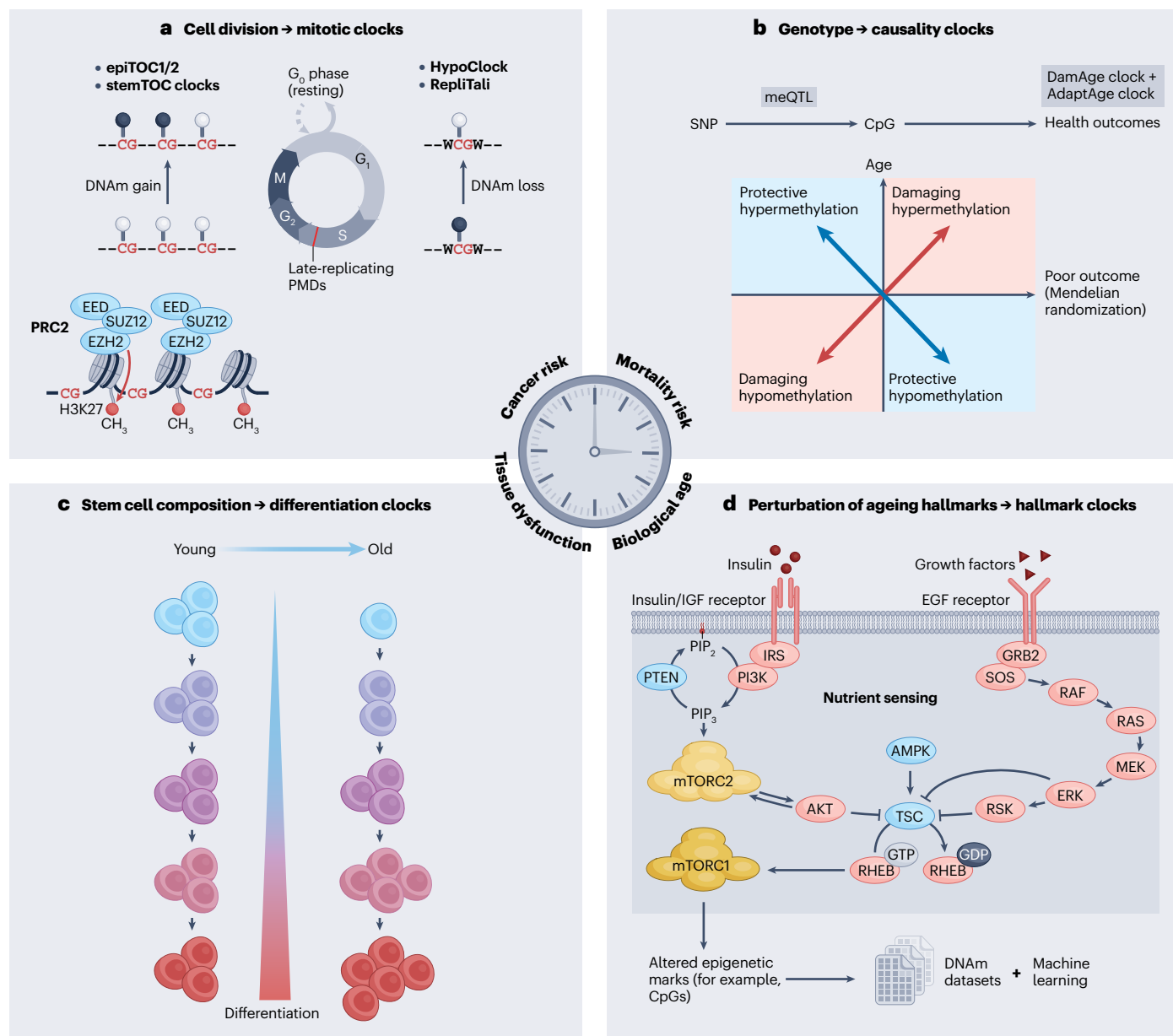


Fig. 4 | Mechanism-specific epigenetic clocks. **a**, DNA methylation (DNAm) loss at solo-CpGs (WCGW with W = A/T) that map to partially methylated domains (PMDs) in late-replicating regions have been used to build one class of mitotic clock. Alternatively, mitotic clocks have been constructed from CpGs universally unmethylated in fetal tissue and marked by the Polycomb-repressive complex 2 (PRC2), which gain DNAm with cell division. These mitotic clocks are universal predictors of cancer risk, consistent with the hypothesis that an increased cumulative number of stem cell divisions is a major cancer risk factor. **b**, Mendelian randomization has been used to identify methylation quantitative trait loci (meQTLs), whereby the underlying CpGs mediate the association of single-nucleotide polymorphisms (SNPs) with health-related traits and outcomes, and are thus potentially causal. Depending on whether the directionality of association of these causal CpGs with health outcome is equal

or opposite to the directionality displayed with age, these CpGs can be divided up into either having an age-related detrimental ‘damaging’ or an adaptive ‘protective’ effect, leading to the construction of the DamAge and AdaptAge clocks, which display corresponding positive and negative associations with all-cause mortality. **c**, Age-related changes in stem cell composition are another key mechanism underlying epigenetic clocks, most clearly illustrated by epigenetic reprogramming to a pluripotent state, which resets many epigenetic clocks to zero. However, tailored differentiation clocks are still to be developed. **d**, In vitro or in vivo experimental perturbation of ageing hallmark pathways, for example, nutrient sensing via the mTOR signalling pathway or modulation of mitochondrial activity, could be used to identify mechanism-specific CpGs for subsequent epigenetic clock construction. Epigenetic clocks tailored to measure these ageing hallmarks are yet to be developed.

versions of the epiTOC1/2 mitotic clocks still display universal age acceleration in cancer and precancerous lesions¹⁶⁵, which suggests that the process by which PRC2 loci acquire DNAm changes upon cell division is largely stochastic, although clearly a stochastic process distinct to the one underlying the Zhang clock. These findings are also consistent with the more homogeneous module composition of the Zhang clock compared with biological ageing clocks such as PhenoAge or GrimAge.

Causal epigenetic clocks

Unravelling the mechanistic components of epigenetic clocks could help elucidate whether age-related DNAm changes have a causal role in ageing and disease predisposition^{5,9,54,167,168}. Although mouse studies have shown that loss of epigenetic information associated with the repair of double-strand breaks is a potential cause of ageing, the specific role of the observed DNAm age acceleration following double-strand break repair is still unclear⁹. Moreover, an *in vitro* study of human radiated cells unexpectedly did not demonstrate epigenetic age acceleration¹⁵¹, whilst other studies indicate that it is only defective DNA repair that dysregulates the epigenome⁵³, in line with a prevailing view that DNA damage is the prime source of ageing⁵⁴. Inference of causal mechanisms in ageing, especially in humans, is very challenging¹⁶⁹. Hence, not unsurprisingly, most efforts to elucidate causality in epigenetic clocks have so far been based on the Mendelian randomization framework¹⁷⁰, which uses genetic variants as instrumental variables for causal inference. A genome-wide association study for epigenetic ageing performed on approximately 9,000 human blood samples revealed hundreds of variants mapping to five genomic loci that modulate age acceleration¹⁷¹, with some of these single-nucleotide polymorphisms overlapping loci that are associated with epigenetic ageing in mice¹⁷². Mendelian randomization also suggested causal influences of menarche, menopause and lipoprotein levels on age acceleration. More recently, a study used Mendelian randomization to identify methylation quantitative trait loci that may mediate the effect of genetic variants on age-related health traits, subsequently building epigenetic clocks from such causality-enriched CpGs¹⁶⁷ (Fig. 4b). By comparing the direction of effect in the Mendelian randomization to the one determined by age, it was observed that causality-enriched CpGs displayed both protective and damaging effects, leading to the construction of two separate 'causality clocks', DamAge and AdaptAge (Table 1 and Fig. 4b), with corresponding clock CpGs displaying age-associated DNAm changes that either promote or are protective of unhealthy ageing¹⁶⁷. Although the nature of the DamAge and AdaptAge clocks were validated in independent cohorts, the causal nature of the underlying CpGs still requires experimental verification.

Experimental elucidation of epigenetic clock mechanisms

Using the Skin&Blood clock⁷⁵ (Table 1), which is a more accurate age predictor than the Horvath clock when applied to cells in culture, it has been shown, through *in vitro* experimentation, that epigenetic ageing is associated with nutrient sensing, mitochondrial activity and stem cell composition but not with other key ageing hallmarks such as senescence, telomere attrition and DNA damage¹⁵¹. For instance, DNAm age from the Skin&Blood clock measured in cultured fibroblasts or endothelial cells transduced with human telomerase reverse transcriptase continues to increase beyond the time point at which senescence occurs in control cells¹⁵¹. Thus, although telomerase reverse transcriptase activation blunts ageing hallmarks such as senescence and telomere shortening¹⁷³, it does not alter the rate of epigenetic ageing

as measured by various epigenetic clocks, including Skin&Blood and Horvath clock. However, these findings do not preclude other clocks, such as PhenoAge or GrimAge, from capturing a senescent process *in vivo*. The finding that DNAm age estimates are strongly influenced by differentiation¹⁵¹ and hence by the relative composition of stem, progenitor and differentiated cells within a tissue, has been nicely validated in the mouse using mouse and pan-mammalian epigenetic clocks¹⁶³, highlighting once again the importance of cell-type heterogeneity, albeit now in the context of individual cell lineages. Thus, an emerging computational challenge is how to estimate relative stem cell, progenitor and differentiated fractions from a bulk tissue DNAm profile that inevitably also includes cell types from unrelated lineages. Estimating such fractions could then help interpret existing clocks or help build novel differentiation-specific clocks (Fig. 4c). To this end, hierarchical cell-type deconvolution algorithms could be developed as conducted, for instance, in the case of solid tissues to estimate fractions of related immune cell subtypes in addition to those of resident non-immune cell types^{118,174}. Experimental perturbation of ageing hallmarks could also offer a means to identify mechanism-specific CpGs, which in conjunction with large DNAm datasets and machine learning could be used to build epigenetic clocks specific to ageing hallmarks (Fig. 4d).

Conclusions and future outlook

The first decade of epigenetic clocks has seen the development of accurate machine learning predictors of chronological age and, to some extent, also of different manifestations of biological age. We have also seen tantalizing evidence that these clocks could be useful for a staggering variety of applications, from predicting the chronological age of endangered beluga whales³⁰ or predicting the future risk of mortality^{17-19,31} and cancer^{34,160}, to predicting the efficacy of health-promoting and anti-ageing interventions in humans and other mammals^{51,52,124}. In the next decade, we envisage that a new generation of epigenetic clocks will be developed, trained on specific and quantifiable forms of biological age, and built in a tissue-type-specific and cell-type-specific manner that allows their potential in ageing, disease prevention and cellular rejuvenation to be fully assessed. To support this endeavour, especially in animal models such as the mouse, for which scalable genome-wide and highly accurate DNAm bead arrays^{27,175-177} and epigenetic clocks^{161,162,178} are now available, it will be paramount to build high-quality pan-tissue DNAm reference atlases at cell-type resolution^{132,175} as well as to improve existing ones in humans^{123,132,179}. In the context of animal studies, where the experimentalist has complete control over genetic and environmental confounders, near-optimal inference could hinge on addressing the confounding posed by cell-type heterogeneity.

Advances in single-cell¹⁸⁰ and cost-effective DNAm assays²³ will usher in the development of clocks derived from single-cell data or cell-type-specific clocks that can be robustly applied at single-cell resolution. As discussed here, a key question will be whether single-cell DNAm data will be useful beyond its obvious advantages of identifying, in a relatively unbiased way, multiple cell types within the same tissue and the ensuing ability to construct DNAm profiles of homogeneous cell populations that could be used, for instance, to build more accurate DNAm reference matrices, enabling improved cell-type deconvolution of large bulk tissue DNAm datasets¹⁸¹. It also remains to be seen whether the apparent biological age variability between single cells of the same cell type is relevant to the quantification of the biological age of a tissue.

Glossary

Age-related differentially methylated cytosines

(age-DMCs). Differentially methylated cytosines associated with age. Also commonly referred to as age-associated differentially methylated positions.

Biological age

A correlative measure of chronological age that is more informative than chronological age of the current and future health status of an individual. An example of a directly quantifiable biological age is mortality risk.

Cell-type deconvolution

In this Review, the application of a statistical algorithm that can infer, from a bulk tissue DNAm profile, the fractions of the underlying cell types in the tissue. Once the fractions are estimated it can also refer to further downstream statistical inferences.

Cell-type heterogeneity

When measuring DNA methylation (DNAm) in a bulk tissue sample, this refers to the fact that the measured DNAm level is a weighted average of the DNAm levels of each cell type in the tissue with the weights being proportional to the cell-type fractions in the tissue.

Cell-type-specific epigenetic clocks

An epigenetic clock that yields a DNAm age value for a specific cell type of a tissue.

Chronological age

Age of an organism as measured by physical time, for example, calendar years from birth.

CTCF-binding sites

CTCF (CCCTC-binding factor) is a DNA-binding protein that regulates the 3D structure and function of a genome. Its binding sites often map to chromatin domain boundaries.

Deep learning

A branch of machine learning that uses non-linear functions in multiple hidden processing layers to learn complex representations of data with multiple levels of abstraction.

Deep neural network

The most common form of deep learning algorithm, involving multiple hidden layers with units (neurons) in each layer taking as input a weighted sum of outputs from the previous layer, and producing an output according to a non-linear activation function.

Differentially variable CpGs

A CpG with a DNAm pattern that differs between phenotypes primarily in terms of its DNAm variance. Phenotypes can be categorical (for example, healthy versus disease) or numerical (for example, age).

DNAm age

The actual estimate of a DNAm-based clock for a given sample that the clock is applied to.

DNA methylation

(DNAm). A covalent modification of DNA that occurs at cytosines, mostly within CpG dinucleotide pairs. DNAm is highly cell-type-specific and in regulatory regions is associated with nearby gene expression.

DNAm reference matrix

A matrix of DNAm values, with rows labelling cell-type-specific marker CpGs or genes (for example, gene promoters) and columns labelling the cell types in a tissue, and which can be used in conjunction with a statistical algorithm to estimate cell-type fractions in a corresponding tissue sample.

Epigenetic clocks

A machine learning algorithm specifying a mathematical rule for combining DNA methylation values of specific CpGs in a sample, resulting in a prediction of the chronological or biological age of the sample.

Explainable artificial intelligence

(XAI). An algorithmic, mathematical tool designed to make a complex machine learning (for example, deep learning) predictor more interpretable. It can be applied after the predictor has been built or it could be part of the learning process.

Graph convolutional neural networks

A form of deep convolutional neural network where the input data has a complex network structure and which, in a supervised context, aims to learn complex predictive representations of the data taking the network topology into account.

Importance scores

In the context of a machine learning algorithm, this refers to a positive weight assigned to each feature (for example, CpG) that allows these features to be ranked in order of importance for the final prediction.

Markov chain

A probabilistic process where a random variable can be in a finite number of states, and where it jumps from one state to another according to a well-defined probability distribution.

Mendelian randomization

A causality inference framework that uses the normal random genetic variation in a human population as anchors to explore whether exposure is causal for an outcome.

Methylation quantitative trait loci

Genetic variants that influence DNAm levels at CpGs. These CpGs could be close (<1Mb) to the single-nucleotide polymorphism (*cis*-methylation quantitative trait loci) or far away, say on a different chromosome (*trans*-methylation quantitative trait loci).

Mitotic clocks

A molecular clock that approximates a form of biological age known as mitotic age, which is the age cumulative count of stem cell divisions in a tissue, which may also include divisions from amplifying progenitor cells.

Molecular ageing clocks

An algorithm trained from any form of molecular omics data that can predict chronological age or any of the various forms of biological age.

Principal components

One-dimensional projections of a data matrix of successive maximal variance that are linearly uncorrelated to each other. These one-dimensional projects have weights defined over genomic features and samples and can be used to predict sample features.

Solo-CpGs

Genomically isolated CpGs (that is, no further CpG within a ± 35 bp window) occurring in a WCGW sequence context (W=A/T) that map preferentially to partially methylated domains and late-replicating regions. They lose DNAm upon cell division.

Tensor

A mathematical array representing a higher dimensional generalization of a data matrix, with numerical entries representing data values across multiple dimensions. For instance, an entry in a tensor could represent the DNAm value of a given CpG in a given cell type of a given individual.

In parallel to these developments, it will become important to delve deeper into the meaning and molecular mechanisms of epigenetic clocks, building novel mechanism-specific clocks or devising novel computational strategies that aid interpretation. Deep learning and XAI tools, which have already been transformative in the construction of non-epigenetic ageing clocks^{182,183}, could help realize these goals, for instance, by incorporating the spatial genomic embedding of DNAm values in the context of prior epigenetic data (for example, chromatin states) from resources such as the International Human Epigenome Consortium^{179,184}. Along these lines, a recent study used an interpretable deep learning unsupervised transformer architecture to build a pre-trained foundation model from over 100,000 DNAm profiles, already showing promise for epigenetic clock construction and other computational tasks such as imputation¹⁸⁵. The recent successful application of a deep recurrent neural network to longitudinal electronic health record data of over 1 million Finnish individuals to fairly accurately estimate the risk of short-term mortality¹⁸³ underscores how a deep learning method could be especially powerful if used in conjunction with longitudinal DNAm data from large cohorts, although current cohort sizes and temporal resolution should increase to make this endeavour fruitful.

Determining the role of other molecular mechanisms that drive epigenetic ageing is also a key prospect. For instance, underexplored mechanisms that likely contribute to DNAm clocks are the circadian rhythm^{58,59,186,187} and notably also somatic mutations⁵⁷ and copy number variants. The construction of somatic mutation clocks¹⁸⁸ as well as elucidating how they relate to existing epigenetic clocks will be particularly important to dissect the relative roles of genetics and epigenetics in biological ageing^{57,160}. Exploring epigenetic clocks based on other types of epigenetic data, including chromatin accessibility¹⁸⁹ or histone modifications¹⁹⁰, could further help interpretation. For instance, an ageing clock trained on over 1,000 histone modification patterns from ENCODE identified many genes that have been directly implicated in ageing biology, including circadian rhythm and neurodegeneration¹⁹⁰. Such clocks could also help to determine the causal role of epigenetic modifications in biological ageing.

Longitudinal prospective studies in human cohorts will be critical to further help elucidate the causal role of a clock, as indicated by a recent study demonstrating how faster longitudinal changes in epigenetic clock age estimates were significantly associated with a higher risk of mortality, independently of baseline epigenetic age and other confounders¹⁹¹. Thus, studies with longitudinal measurements in both DNAm and clinical ageing biomarkers should further help quantify the true pace of ageing of individuals, overcoming the current limitations of cross-sectional studies (that is, DNAm measured at one time point only)⁴⁵.

Interestingly, further insights into a potential causal role of DNAm and DNAm clocks in ageing could come from pan-mammalian studies that have uncovered a surprising inverse relationship between the average rate of methylation change in a species with its maximum lifespan, which intriguingly is dependent on chromatin context, being specific to bivalent PRC2 domains²⁸. Finally, there will be a growing need to integrate epigenetic clocks across cell types within a tissue or across different tissues, since the biological ageing of cell types within a tissue or of different tissues in an organism does not proceed in isolation but via a complex system of intercellular and inter-organ interactions^{134,192}. In addition, computational integration of epigenetic clocks with other molecular ageing clocks (for example, transcriptomic^{149,193,194}, proteomic^{195,196} or metabolomic¹⁹⁷ clocks) may

be critical as different molecular layers can complement each other, optimally capturing different forms of biological age that together determine the physiological state of an organism⁶.

The future construction of single-cell, cell-type-specific and mechanism-specific clocks will encounter many statistical and computational challenges. It is hoped that the methods and guidelines provided here on how to overcome some of these specific computational challenges will be useful for the community and help us towards the goal of extending a healthy lifespan.

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Author contributions

A.E.T. wrote the article. Both authors researched the literature and reviewed and edited the manuscript.

Competing interests

S.H. and his team developed the first epigenetic clocks for human saliva, for all tissues (pan-tissue clock), for human mortality risk prediction (PhenoAge, GrimAge), and pan-mammalian clocks. The Regents of the University of California are the sole owner of patents and patent applications directed at epigenetic biomarkers for which S.H. is a named inventor; S.H. is a founder and paid consultant of the non-profit Epigenetic Clock Development Foundation that licenses these patents. S.H. is a principal investigator at the Altos Labs, Cambridge Institute of Science, a biomedical company that works on biological rejuvenation. A.E.T. declares no competing interests.

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